Substituting Reference σ-Profiles Degrades Solubility Prediction and Separates COSMO-SAC Error from Input Error

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Abstract

Physics-informed models route predictions through interpretable physical quantities, expecting that accurate reference values for them improve accuracy. We measure that for solid–liquid solubility. A graph neural network predicts a molecule's charge-density profile, the fixed thermodynamic model COSMO-SAC turns a pair of profiles into an activity coefficient and hence a solubility, and a quantum-chemical database tabulates the same profiles. Grounding names two opposite operations on it, with opposite signs: supervising the predicted profile against it during training improves solubility accuracy over three seeds by 0.20 in mean absolute error; substituting the reference for the learned profile at inference degrades it by 0.41. It is also an instrument: on infinite-dilution activity data it bounds the model's own error from below and its inputs' from above. The bound on the inputs barely moves between chemical strata while the total error differs across them by more than an order of magnitude, so the comparison is a map over 59 strata, not one verdict. Under one admissibility rule applied alike to every stratum, fifteen are large enough to bound and six keep their margin's sign under every publication deletion; those six are three row sets, and on one—the glycol-ether solvents, 182 measurements over 43 molecule pairs—the model's error is the larger term, by +2.04 (90% interval $[+1.27, +2.83]$). The map reports the rest without establishing it. Trained through the misspecified model, the learned profile departs from its reference along a shared direction. Grounded latents are not reliably interpretable when the model consuming them is misspecified.

1 Introduction

How much of a solid dissolves in a given liquid decides whether a drug can be formulated as a tablet or an injection, which solvent a crystallisation is run in, and how a product is purified. It must be measured one solute, one solvent and one temperature at a time, so most industrially relevant combinations have never been determined, and independent determinations of the same system disagree by enough to floor what any predictor can achieve. Dissolving a solid also trades the cost of breaking up a crystal against the cost of mixing the freed molecules into the solvent, so a predictor has to be right about two unrelated kinds of chemistry at once, on molecules unlike any it was shown.

Predictors divide by how much physical structure they impose. Direct models regress solubility from descriptors or from a representation learned off the molecular graph; FastSolv[1] does so with a deep neural network and leads on extrapolation to new solutes. They are the more accurate there and they are opaque: the prediction arrives as a single number, with no account of which part of the chemistry it captured. Physics-informed models instead predict the thermodynamic quantities the physical picture names—an activity coefficient for the solute–solvent interaction, a melting temperature and heat of fusion for the crystal—and combine them through a fixed equation, as SolProp[2] does with a thermodynamic cycle of separately learned components. What that buys is decomposability: each part is a quantity chemists tabulate on its own, so it can be supervised against reference data a direct regressor has no way to use. On accuracy the black box leads and the physics-informed cycle trails it. Learned surrogates for the activity coefficient have advanced quickly, and split the same way. Winter et al.[3] predict $\ln \gamma_2^{\infty}$ from SMILES with a language model and Sanchez-Medina et al.[4] embed a Gibbs–Helmholtz constraint in a graph network for its temperature dependence, both making the activity coefficient the regression target; a second line instead keeps a mechanistic model and learns its input. COSMO-SAC[5] is a segment-activity reformulation of Klamt's COSMO-RS, and the σ-profile it consumes—screening-charge density over a molecule's cavity—is that framework's construct, tabulated from quantum chemistry.[6] Predicting it from structure removes that calculation from the pipeline; TeNNet-SAC[7] does so while also learning the segment-activity kernel, and gains from truer profiles.

When a fixed physical structure improves or degrades a learned model is the broader question of physics-informed machine learning,[8] whose dominant forms impose structure as a differentiable object: a governing equation enforced through the loss,[9] gray-box hybrids composing learned components with fixed mechanistic maps,[10] and operator learning with a trainable correction to a misspecified equation.[11] The prevailing expectation is that more physics improves generalisation, yet the same structure has documented failure modes in which it makes optimisation or accuracy worse:[12] in neural PDE solvers, a learnable term added to a fixed discretised operator absorbs truncation artifacts rather than physics, so the apparent interpretability is illusory[13]—though that error source is numerical and has no external reference. The same concern appears in concept-bottleneck models as *concept leakage*, a learned concept silently encoding information beyond its nominal meaning.[14–16] Recent work traces one route to leakage to the downstream model: a misspecified head forces the encoder to deform the concept into carrying a dependence the head cannot represent,[17] and overwriting a frozen head's leaked concepts with ground truth can then degrade accuracy.[18] The gap between a fixed physical map and reality is the model-discrepancy problem of Kennedy and O'Hagan[19] and its calibration critique;[20] four further literatures supply pieces used below—forecast decomposition, which splits a fixed predictor's error as §2.1.1 does; quasi-maximum likelihood, which gives the pseudo-true limit a misspecified closure converges to; identifiability, for when a composed model's parts are separable at all; and the validity of composed pipelines, for what a bound on one component licenses about the whole (§S12).

The design rests on an assumption rarely stated because it is taken for self-evident: a model supplied with physical inputs closer to their true values predicts better. For one class of intermediate that can be tested outright, because a reference value is tabulated and can be handed to the model in place of the one it computed. It needs testing, because when a predictor of this shape is inaccurate either the fixed physical model is wrong or the learned quantities feeding it are uninformative, and the two call for opposite remedies. What this literature lacks is a way to *measure*, rather than diagnose, which of the two limits a composed predictor's accuracy, and whether the learned intermediate has stopped reporting the quantity it is named after; in the concept-bottleneck case no external reference exists against which either question could be settled. This work supplies one—a reference σ-profile for solubility, tabulated substituent constants for a second chemical domain—and holds the mechanistic kernel fixed, so the tabulated value stays available as an external reference for the learned one.

Here a graph neural network predicts the pair of charge-density distributions COSMO-SAC consumes, in place of the quantum-chemical calculation that ordinarily supplies them. The tabulated counterpart can enter at either of two points: as a target the network is trained toward, or as a replacement for what the network computed at the moment of prediction (Fig. 1). The thermodynamic model is differentiable, so both routes are open on a single trained network, and the tabulated values are put to a third use, dividing the fixed model's error between the map and the inputs handed to it. Only one route helps, and the reason is a property of the fixed map. Solubility data cannot say which component is at fault, one measurement constraining the pair jointly; activity coefficients can, because both molecules carry a tabulated distribution and the two error terms separate. Where they separate, an ordering decides the remedy: a map whose own error stands above what its inputs fail to carry does not become more accurate when handed a better input, but more exposed, the accurate input no longer masking the bias the map introduces. That ordering is a property of the chemistry the map is asked about, so the instrument returns a map over chemical strata rather than a verdict. Two consequences follow. A learned intermediate trained through a misspecified map cannot be read as the physical quantity it is named after: the predicted distribution drifts away from the tabulated one along a direction shared across molecules. And substituting a reference value for it at prediction time is not a safe improvement but a change of regime, which is why the substitution here costs more than the supervision gained. Neither consequence is peculiar to charge distributions: tabulated substituent constants behave the same way inside a fixed Hammett relation for acid dissociation, improving the aromatic series that relation was built for and degrading the ones outside it.

2 Computational methods

2.1 The composed predictor and its error split

For a solute s in solvent v at temperature T, the target is $y = \ln x_2$, the log saturated mole fraction. Dissolving a solid trades breaking the crystal lattice (Φ) against mixing the freed molecules into the solvent ($\ln\gamma_2$), and solid–liquid equilibrium (SLE) sets the log-solubility to their negative sum:

$$\ln x_2 = -\underbrace{\frac{\Delta H_{\text{fus}}}{R}\Big(\tfrac{1}{T} - \tfrac{1}{T_m}\Big)}_{\Phi(T)\ \text{(crystal)}} - \underbrace{\ln\gamma_2(x_2, T)}_{\text{activity}}, \quad (1)$$

with the constant-ΔC_p correction omitted for clarity. The activity term is produced by a *differentiable closure* g: a graph encoder predicts a σ-profile—the distribution of screening charge density over a molecule, the histogram a COSMO model consumes—for each molecule, and COSMO-SAC,[5] a segment-activity reformulation of COSMO-RS, maps the pair of profiles to $\ln\gamma_2$. The segment-interaction energy inside that map—the electrostatic-misfit and hydrogen-bonding terms setting what two surface segments cost each other—is the closure's *kernel*, the part the published revisions rewrite, so a closure variant is named for the year of its kernel: the deployed *2002 kernel*, and the *2010 kernel* that types segments as hydrogen-bond donors or acceptors instead of treating hydrogen bonding with one parameter. Write $z^\star$ for the *reference* physical latent, the σ-profile tabulated in the VT-2005 database,[6] and $\hat{z}$ for the head's own prediction. Feeding $z^\star$ in place of $\hat{z}$ (a σ-oracle) isolates the closure from encoder error: if the closure were well specified, $g(z^\star)$ should be the best available activity estimate. Each trained configuration, and each closure variant evaluated, is an *arm*; a *Direct-GNN* control arm drops the closure and emits $\ln x_2$ directly, and the two were tuned separately (§2.4).

Everything below is stated for a general *composed predictor* $\hat{y} = g(h(x))$ in which a fixed closure g consumes a learned physical interme-

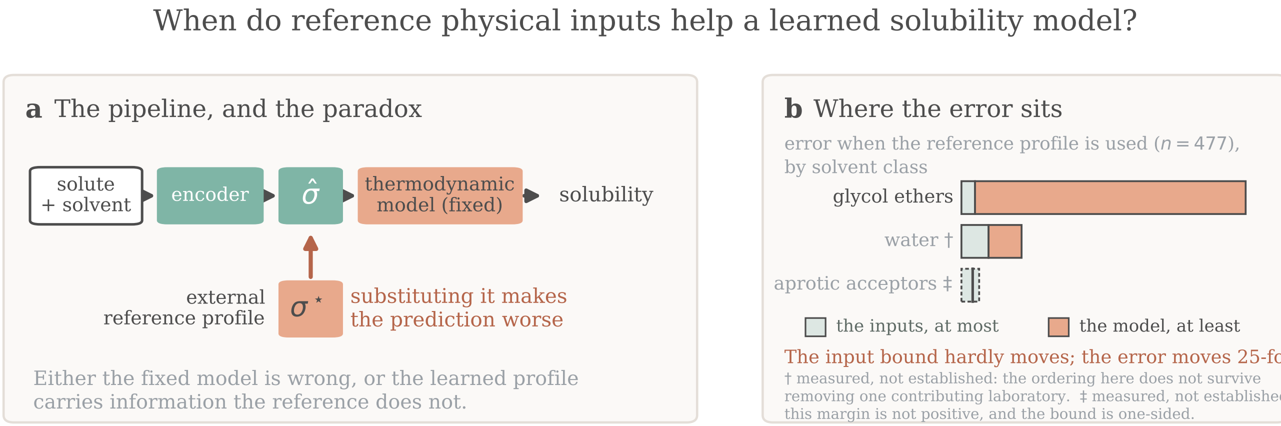

Figure 1: **Overview.** **(a)** A learned encoder predicts a charge-density profile $\hat{\sigma}$ that a fixed thermodynamic model turns into a solubility; the arrow marks the evaluation-time substitution of the reference $\sigma^\star$, which makes it worse. **(b)** The counterpart error on infinite-dilution activity data, by solvent class; only the unmarked class is established, each mark carrying its own reason, and each block is a one-sided bound rather than an interval. Training-time supervision is not drawn.

diate $z = h(x)$, against a matched free predictor that shares the encoder and drops g; solubility is one instance and the pKa closure of §3.5 a second. Nothing in Lemma 1 is specific to COSMO-SAC.

The crystal/activity split of Eq. (1) is not identifiable from solubility alone. At infinite dilution with $\Delta C_p = 0$ the observable is affine in $1/T$, and the activity intercept and slope already span that plane, so the profiled Fisher information for ΔH_{fus} is exactly zero and the indeterminacy set is two-dimensional; at finite dilution the composition factor $(1 - x_2)^2$ leaves an information of order x_2^2. The deficiency closes only under a rank-completing external constraint—a direct crystal label, or a fixed activity slope (Lemma S1, §S2). That is why an under-determined factor needs external single-component data and why compensation between the factors is expected; it says nothing by itself about whether grounding helps.

2.1.1 The closure–insufficiency decomposition

Let m be the activity target—the experimental infinite-dilution activity coefficient $\ln \gamma_2^\infty$—and $g(z^\star)$ the closure evaluated on the reference latent.

Lemma 1 (Exact closure–insufficiency decomposition and its one-sided bounds)**.** *For a* fixed *(non-fitted) closure g, with $z^\star$ denoting the closure's* entire *argument (the profile pair together with the temperature wherever the closure reads one),*

$$\underbrace{\mathbb{E}\big[(m - g(z^\star))^2\big]}_{\textit{ref.-input MSE}} = \underbrace{\mathbb{E}\big[\mathrm{Var}(m \mid z^\star)\big]}_{B_{\mathrm{insuff}}\ (\textit{irreducible})} + \underbrace{\mathbb{E}\big[(\mathbb{E}[m \mid z^\star] - g(z^\star))^2\big]}_{B_{\mathrm{closure}}\ (\textit{decoder bias})}, \tag{2}$$

the cross term vanishing identically. Moreover $B_{\mathrm{closure}} \geq (\mathbb{E}[m] - \mathbb{E}[g])^2$, and $B_{\mathrm{insuff}} \leq \mathbb{E}[\mathrm{Var}(m \mid \mathrm{bin}(g(z^\star)))]$ for any binning of $g(z^\star)$.

Proof. (a) The identity, and the cross-term condition. $g(z^\star)$ is $\sigma(z^\star)$-measurable by construction, since $z^\star$ is the closure's whole argument. Split the residual as $m - g(z^\star) = [m - \mathbb{E}(m \mid z^\star)] + [\mathbb{E}(m \mid z^\star) - g(z^\star)]$, square and take expectations; the squared terms are B_{insuff} and B_{closure}, and the cross term vanishes by the tower rule, $\mathbb{E}[(m - \mathbb{E}(m \mid z^\star))(\mathbb{E}(m \mid z^\star) - g(z^\star))] = \mathbb{E}[(\mathbb{E}(m \mid z^\star) - g(z^\star))\,\mathbb{E}(m - \mathbb{E}(m \mid z^\star) \mid z^\star)] = 0$, because the second factor is $\sigma(z^\star)$-measurable. **That measurability is the condition, and it fails if any argument of g is left out of the conditioning variable**: with $g = g(z^\star, T)$ and T

excluded, the cross term need not vanish and the binning in (c) is no longer a coarsening of the conditioning σ-algebra, so neither the identity nor the upper bound survives. Where g reads a temperature—the broad IDAC set of §3.2.1, dim $=$ 103—we accordingly take $z^\star = (\text{profile pair}, T)$; on the VT-2005-matched set every row sits at 298 K and the two readings coincide. *(b) Jensen lower bound.* With $\Delta(z^\star) = \mathbb{E}(m \mid z^\star) - g(z^\star)$, $B_{\text{closure}} = \mathbb{E}[\Delta^2] \geq (\mathbb{E}[\Delta])^2 = (\mathbb{E}[m] - \mathbb{E}[g])^2$, using $\mathbb{E}[\mathbb{E}(m \mid z^\star)] = \mathbb{E}[m]$. It needs *no* smoothness assumption; its one requirement is that m and g share the $\ln \gamma_2^\infty$ reference state, which the closure validation of §2.3 establishes. *(c) Law-of-total-variance upper bound.* For any binning $B = \text{bin}(g(z^\star))$ one has $\sigma(B) \subseteq \sigma(z^\star)$, and conditioning on a coarser σ-algebra cannot reduce expected conditional variance: $\mathbb{E}[\text{Var}(m \mid B)] \geq \mathbb{E}[\text{Var}(m \mid z^\star)] = B_{\text{insuff}}$. *(d) The estimand is convention-free; the bound is not.* $B_{\text{insuff}} = \mathbb{E}[\text{Var}(m \mid z^\star)]$ is a functional of the law of $(m, z^\star)$ alone and does not involve g, so it is identical across closure conventions and any $\text{MSE}(g_1) - \text{MSE}(g_2)$ is entirely $B_{\text{closure}}(g_1) - B_{\text{closure}}(g_2)$. The bound of (c) is not convention-independent, and no contradiction arises: two conventions coarsen the same $\sigma(z^\star)$ along two different statistics, giving two valid upper bounds on one quantity, of which the minimum is again valid but is biased low in finite samples, so we report it and do not headline it. □

B_{closure} is the systematic discrepancy of a fixed physical map in the sense of Kennedy–O'Hagan,[19,20] and its dominance over B_{insuff} is the empirical content of the claim that the ceiling is set by the closure rather than by the inputs. Where model-discrepancy calibration *fits* a correction from held-out observations, here g is fixed and $z^\star$ externally observed, so the map's error is *attributed* rather than corrected—available only where an external latent oracle exists. **Exactness is contingent on g being fixed**: for a *fitted* decoder the cross term does not vanish and a plug-in point split is invalid. Because B_{insuff} is bounded above and B_{closure} below, a positive *separation margin* $\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$ establishes that the closure's own error exceeds what input insufficiency can account for, while a non-positive one is a failure to separate and licenses nothing: a reversal would need a *lower* bound on B_{insuff}, which this instrument does not supply.

What population, and what sets the resolution. The population is the *measurement superpopulation*: a draw is a nominal compound pair *at a temperature*, together with the conformer, tautomer and protonation state actually populated there, and an independent determination of $\ln \gamma_2^\infty$ for it. Neither the conformational state nor the replicate is part of the conditioning variable, so $\text{Var}(m \mid z^\star, T)$ is the spread of $\ln \gamma_2^\infty$ across the conformational ensemble and across replicate determinations at one reference profile and temperature—nonzero for physical reasons unrelated to sample size. Experimental label noise is a *component* of it, not a term beside it: an independent zero-mean error on m enters $\text{Var}(m \mid z^\star, T)$ directly, so it sits inside every bound reported here and is never added on. What the design cannot do is *observe* it, neither set holding two rows at a common argument. The bounds are therefore conservative in one direction only—were the superpopulation variance negligible the finding would be the stronger $B_{\text{closure}} = \text{MSE}$—so nothing reported can be an artifact of over-estimating B_{insuff}, no entry anywhere is the share of the error the inputs contribute, and a low cluster-bootstrap frequency P is loss of resolution and never evidence for the inputs. Resolution is governed by $\dim(z^\star)/n$.

2.1.2 The decomposition and the grounding gap

The decomposition is the measurement the conclusions rest on: assumption-light, needing neither a representable encoder nor a lossless bottleneck. Alongside it we report a second, purely observable quantity, the *grounding gap* $\mathcal{G} := R_{\text{orc}} - R_{\text{e2e}}$, the oracle risk minus the best end-to-end risk; the paradox is $\mathcal{G} > 0$. Under a representable encoder (A1) and a lossless bottleneck (A2) it is sandwiched $0 \leq \mathcal{G} \leq B_{\text{closure}}$, so $\mathcal{G} > 0$ would certify $B_{\text{closure}} > 0$ (Theorem S1,

§S2). A1 is empirically doubtful—the encoder is transfer-limited, a linear probe dropping from train $R^2 \approx 0.76$ to test $R^2 \approx 0.45$—and A2 is unverifiable and physically dubious; when A2 fails, $\mathcal{G} > 0$ conflates closure misspecification with input insufficiency, a well-specified closure with a lossy latent giving $B_{\text{closure}} = 0$ yet $\mathcal{G} > 0$. We therefore read $\mathcal{G}$ as corroborating evidence and base the model-dominance conclusion on the decomposition.

The sign rule, and scope. Whether the *trained* physics model outperforms the *trained* black box at sample size n follows a sign rule. Writing each trained risk as an asymptote plus estimation variance, $\mathbb{E}[\hat{R}_{\text{phys}}(n)] \approx R_{\text{e2e}} + V_{\text{phys}}(n)$ and $\mathbb{E}[\hat{R}_{\text{direct}}(n)] \approx R_{\text{direct},\infty} + V_{\text{direct}}(n)$ with $V(n) \downarrow 0$, the *physics penalty at budget* n is $\mathcal{T}(n) \approx \Delta_\infty + [V_{\text{phys}}(n) - V_{\text{direct}}(n)]$, with $\Delta_\infty := R_{\text{e2e}} - R_{\text{direct},\infty}$ its *asymptotic* part.

Proposition 1 (Sign of the physics penalty)**.** *(i)* $\mathcal{T}(n) < 0$ *(physics helps) iff* $V_{\text{direct}}(n) - V_{\text{phys}}(n) > \Delta_\infty$*: the variance a constrained prior saves must exceed its asymptotic bias. (ii) If* $\Delta_\infty > 0$ *and the variance gap vanishes* ($V_{\text{direct}} - V_{\text{phys}} \to 0$)*, there is a critical budget* $n^\star$ *beyond which* $\mathcal{T}(n) > 0$ *for all* $n > n^\star$*.*

Both halves are immediate from $\mathcal{T}(n) = \Delta_\infty - D(n)$ with $D := V_{\text{direct}} - V_{\text{phys}}$ (proof, and the limiting refinement in which $\Delta_\infty = B_{\text{closure}} - \mathcal{G}$ when the control is asymptotically Bayes, in §S2). The consequence used below is that a prior is predicted to hurt precisely in the mature-data, low-noise, well-covered regime. Each physics-versus-control difference reported here is a penalty at the budget actually trained, not an asymptotic one; there is no "optimal grounding" theorem, and the decomposition does not improve out-of-distribution accuracy. Its variance term $D(n)$ is what carries the semiparametric phenomenon in which an estimated propensity outperforms the known-true one:[21,22] a fitted nuisance can be the more *efficient* estimator even when its bias is no smaller, so a true-valued input need not give the lower total risk at finite n.

2.2 Data, splits, and labels

Solubility measurements come from BigSolDB 2.0:[23] 127,930 rows over 15,665 solutes and 221 solvents between 243 and 438 K, partitioned by Bemis–Murcko scaffold into 111,724 training, 8,103 validation and 8,103 test rows so that every test solute is unseen at training; a melting-point-independent miscibility/structure filter sets row membership, and crystal T_m, ΔH_{fus}, Hansen parameters and infinite-dilution activity coefficients $\ln \gamma_2^\infty$ are merged on as per-solute labels where available. Crystal supervision is scarce—the test and validation splits contain essentially no measured ΔH_{fus}, the quantitative basis of the identifiability argument (§2.1). The external single-component data used for grounding are an open crystal-property pool (~15,000 melting points), the VT-2005 σ-profile database, and ThermoML infinite-dilution activity coefficients.[24] Two hazards the pipeline handles explicitly—melting-point unit detection, and the instability of the seeded scaffold split across pipeline versions—are recorded in §S9.

The label-noise floor, and the two functionals it is read with. Independent determinations of the same system disagree, and that disagreement floors what any predictor can achieve. On the raw BigSolDB rows, group by (solute, solvent, temperature to 1 K) and keep the groups reporting at least two distinct sources; within a group the disagreement statistic is the mean absolute deviation of $\ln x_2$ from the group mean. Aggregating that per-group statistic over groups requires a choice of functional, and the two standard ones differ by a factor of five, so both are reported: the mean over groups is 0.15 $\ln x_2$ units for the 3948 groups backed by two sources and 0.31 for the 349 backed by three, while the median over the same groups is 0.03 and 0.09. Collapsing exact-duplicate values first—the same measurement re-tabulated across compilations, which would otherwise deflate the floor—leaves the mean at 0.16 over 4261 groups. The scaffold-split MAE reported here (1.70 to 1.85) is above the larger functional by a factor of five, so label noise does

Table 1: Terminology, and the two symbols that would otherwise collide. Each term is defined here and used below without re-glossing.

Term	Meaning in this manuscript
closure g	the fixed, non-fitted physical map from the intermediate to the observable: here COSMO-SAC, in §3.5 a Hammett relation
kernel	the segment-interaction energy inside COSMO-SAC; a closure variant is named for its kernel's year (*2002*, *2010*)
arm	one trained configuration, or one closure variant evaluated; *DirectGNN* is the closure-free control arm
$z^\star$, $\hat{z}$	the reference (tabulated) latent and the network's own prediction of it; a *σ-oracle* feeds $z^\star$ where $\hat{z}$ would go
grounding	two opposite operations on one reference database: *supervising* $\hat{z}$ against it during training, and *substituting* $z^\star$ for $\hat{z}$ at evaluation
B_{closure}, B_{insuff}	the closure's own error, and what its inputs cannot resolve, Eq. (2); B_{closure} is bounded below and B_{insuff} above, so the comparison is one-sided
margin	$\text{MSE} - 2B^{\text{up}}_{\text{insuff}}$; positive establishes $B_{\text{closure}} > B_{\text{insuff}}$, non-positive is failure to separate and licenses nothing
$\mathcal{G}$	the grounding gap $R_{\text{orc}} - R_{\text{e2e}}$, renamed from Γ so as not to collide with the segment activity coefficient Γ_S
stratum, map	a chemically defined row set of the broad IDAC set, and the 59 of them the estimator is run on
boundable	$n \geq 40$, so the fixed cell's eight bins hold five rows each
admissible	boundable *and* sign-stable under deletion of each contributing publication, in all four unit $\times$ convention cells (§2.5)
full, deployed	the combinatorial convention: *full* carries the Staverman–Guggenheim term, *deployed* is residual-only and is what this pipeline runs
broad IDAC set	477 IDAC $\cap$ UD rows, 185 pairs, all temperatures, UD profiles
VT-2005-matched set	60 pairs at 298 K carrying a VT-2005 profile on both sides
σ	alone, the screening-charge-density profile; $\sigma(\cdot)$ with an argument, the generated σ-algebra; σ_{H}, the Hammett substituent constant of §3.5; σ_η, experimental label noise

not account for the accuracy ceiling under either reading. This floor is a mean absolute deviation in $\ln x_2$; the aleatoric estimate published for this task, an inter-laboratory RMSE of 0.75 in $\log_{10} S$ on 34 duplicate solutions,[1] is a different functional on a different base, and the two are neither pooled nor converted into one another anywhere below.

Disclosure: the scaffold guarantee is not certified for these runs. The released stream builder excludes any pool molecule whose scaffold appears in validation or test and aborts on a missing split file—but that is a property of the builder, not a certified property of the runs reported here. One σ-stream build on disk was generated against the wrong split files, 91 of its 1425 rows being held-out solutes by canonical SMILES, and per-run stream provenance was not logged, so the artifacts cannot certify which build fed which arm. §S9 gives the file-order evidence, the three facts that bound what a leak could do, and the one molecule it touches in the n=44 analysis of §3.3. We therefore do not assert the scaffold guarantee for these runs.

2.3 Model, closure, and training

A *shared* message-passing encoder embeds solute and solvent, a bipartite cross-attention block lets them interact, and the resulting pair representation feeds every head; two interchangeable encoder alternatives do not outperform it. A fusion head predicts T_m and ΔH_{fus}, giving the ideal term $\Phi(T)$ of Eq. (1). The activity term $\ln \gamma_2$ comes from one of three differentiable closures. Two are controls, the fitted NRTL model and its symmetric one-parameter collapse. The third, parameter-free, is the closure under study: a σ-profile head emitting a 51-bin screening-charge-density profile per molecule, and a COSMO-SAC layer mapping the profile pair to $\ln\gamma_2$ with an optional Staverman–Guggenheim combinatorial term—the *full* versus *deployed residual-only* distinction on which §2.5 and §3.2 turn. The saturated mole fraction solves $\ln x_2 = -\Phi(T) - \ln\gamma_2(x_2, T)$ as a damped fixed point in x_2; a bounded, gated adaptive correction perturbs the physical parameters rather than the output, so the physical decomposition survives it.

External single-component data enter through *grounding streams*—the training-time sense of grounding. Each stream is a sidecar of *self-solvent* rows (solvent = solute, no solubility label, only the target factor's mask active), interleaved into training under the guard above; the streams ground the crystal factor from the melting-point pool and the activity factor from the σ-profile database, identifying from external data a factor that solubility alone leaves non-identifiable and that a black-box predictor cannot use. Training runs a three-stage curriculum under a weighted loss: a warm-up on the auxiliary streams alone, then joint training against solubility, then a low-rate stabilisation stage. The schedule pinned for the arms of Fig. 2 freezes the σ head at the end of the warm-up, so the σ stream enters the warm-up only.

The in-house closures, and where their validation stops. The dispersion-off closures in the fidelity measurement (§3.2.3) are our own differentiable implementations, so those arms share one code path, and each consumes its own model-prescribed σ-averaging (Mullins for 2002, Hsieh for 2010). Our 2002 layer—the *fair 2002* arm wherever the kernels are compared, because it reads all three profile channels—reproduces the NIST reference COSMO-SAC-2002[25] to RMSE 0.0035 $\ln\gamma$ on identical VT-2005 profiles, and our COSMO-SAC-2010/dsp layer,[26,27] built on the *typed* latent of three donor/acceptor-resolved σ-profiles per molecule in place of one, reproduces the NIST 2010 reference with dispersion *off*. The dispersion term is *not* validated to that standard, which is why every dispersion-off contrast is computed on our layers while the *2010, dispersion on* runs are quoted from `cCOSMO`.

The same closure against a published evaluation. Reproducing a reference implementation fixes the code; it does not say whether the error this paper decomposes is the error a published COSMO-SAC has, or something peculiar to this data assembly. Table 2 answers that. De Souza et al.[28] score the original Lin–Sandler COSMO-SAC on the infinite-dilution activity-coefficient database distributed with *The Properties of Gases and Liquids*, at AAD $= \mathbb{E}|\ln\gamma^\infty_{\text{calc}} - \ln\gamma^\infty_{\text{exp}}|$ of 1.75 with HF-TZVP profiles and 0.68 with BP-TZVPD-FINE: one closure, two profile databases, a factor of 2.6 between them. We ran the deployed `CosmoSacLayer`—its own class, constants and damped fixed point—over the same three record files, resolving components through the repository's CAS-to-InChIKey map and matching by exact InChIKey to this paper's VT-2005 artifact: 7889 of 8138 usable records (96.9%), 2279 distinct pairs, each at its own temperature. It lands inside the bracket the published closure spans across profile databases, on both statistics and under both conventions, and so do this paper's own two sets. Two limits stay open: the profile database is the one axis we cannot match, VT-2005 being what this paper deploys and the axis that publication shows dominates; and the publication's 6977 points are not proven to be the same rows as the 7889 scored here, the record selection not being stated file by file. Subject to those, the closure error decomposed in §3.2 is the error a published COSMO-SAC has, and the structure inside the published set is the same H-bond-kernel localisation reported in §3.2.4, on 130 times the data and someone else's compilation.

2.4 Evaluation protocol, and the DirectGNN control

The *oracle substitution* replaces the predicted σ-profile $\hat{z}$ with the reference VT-2005 profile $z^\star$ before the closure, matched by canonical SMILES. It is applied three ways to rule

Table 2: The deployed closure against a published evaluation of the same closure. AAD is the mean absolute error in $\ln \gamma^{\infty}$.

COSMO-SAC-2002 on	profiles	n	AAD	R^2
As published[28]				
PGL 6th ed. IDAC set	HF-TZVP	6977	1.75	0.88
PGL 6th ed. IDAC set	BP-TZVPD-FINE	6977	0.68	0.96
The same set, through this paper's closure				
full convention[a]	VT-2005	7889	0.98	0.88
residual (deployed)	VT-2005	7889	0.77	0.92
non-aqueous rows	VT-2005	4347	0.47	0.68
aqueous rows	VT-2005	3542	1.13	0.83
This paper's own sets, residual convention				
broad IDAC set	UD	477	1.12	0.61
VT-2005-matched set	VT-2005	60	0.81	0.40

[a] The like-for-like row, a published COSMO-SAC carrying its Staverman–Guggenheim term; the residual row is the convention this paper's B_{closure} is measured in. Worst cell: water in an organic solvent, AAD 2.02 with a +1.92 bias on 468 rows. At 298 K and non-aqueous—the regime the n=60 set occupies—AAD is 0.42 on 1033 rows, against that set's 0.81, so this paper's set is if anything the harder one.

out implementation artifacts: at evaluation only; injected at training time so the model co-adapts; and as a channel-swap control, injected at training under a coordinate-descent schedule that freezes the crystal branch during the SLE phase, so that only the activity branch refits against it. The last two are at seed 42 only. An analogous override substitutes reference crystal targets. The activity target for the decomposition is the infinite-dilution $\ln \gamma_2^{\infty} = \lim_{x_2 \to 0} \ln \gamma_2$, on two sets that differ in their reference database and their regime (§3.2.1–§3.2.2).

DirectGNN shares the encoder, interaction, readout and pair representation, adds a thermometer (cumulative-threshold) temperature encoding, and maps directly to $\ln x_2$ with no heads, closure or solver. The two arms therefore share their representation but not their inputs or their tuning: the control receives an explicit temperature encoding the physics arm does not, seeing T instead through $\Phi(T)$ and the closure, and the optimisation settings were tuned per arm and differ by up to an order of magnitude. The TGNNSolv−DirectGNN gap is accordingly a comparison of two separately tuned pipelines that differ in the thermodynamic bottleneck, not an isolation of the bottleneck, and every physics-penalty comparison below carries that caveat (§3.6).

The controlled comparisons of §3.1 are run at three seeds and reported as mean±sd on a cross-arm row set; 5608 is that lock, the test rows supervised and finite in every arm, so all arms score identical rows. Table 3 names which run family produced which numbers, how many seeds it carries, and the script and artifact each regenerates from.

2.5 Estimating the split: one cell, one convention, one rule

Four analyst choices set the law-of-total-variance bound—the bin count, where the equal-count bin edges fall, the within-bin variance convention, and the combinatorial convention—and all four are fixed here, before any margin is reported.

The estimator cell. Eight equal-count bins of $g(z^{\star})$, the *unbiased* within-bin variance, and the row unit, applied identically to every set and every stratum whatever its size. Enumerating every equal-count contiguous partition, the bin-edge choice is immaterial on the broad IDAC set and material on the VT-2005-matched set, whose margin changes sign inside the envelope; we accordingly read that set's margin as consistent with zero and carry the claim on the broad set in stratified form.

The convention rule. The margin depends on the combinatorial convention, so one rule governs both sets: headline the *deployed* residual-only convention, pairing MSE and the bound computed under that same convention. Headlining the deployed convention on the smaller set and the full convention on the broad one would be the favourable choice on each set taken separately; one rule applied to both costs the broad set's margin about half its value. The Jensen constant-offset bound is convention-specific—it separates under the full convention and inverts under the deployed default—so what is reported rests on the model-free law-of-total-variance bound, which fits nothing and so lets no held-out row borrow strength from a near-twin. Both bounds are valid on the one population quantity, by Lemma 1(d), and so is the smaller.

The stratification rule. The broad set is strati-

Table 3: Run families: which runs produced which numbers, and at how many seeds. Scripts are in `scripts/analysis/`, artifacts under `results/`.

Run family (script)	Seeds	What it produced
σ-grounding arms, GPU, full corpus (`run_e5_comparison`)	$42, 43, 44$, mean$\pm$sd on the n=5608 lock	Figs. 2 and 3, Table S2, the $+0.41$ and -0.20 of §3.1
Co-adaptation and channel-swap oracle controls; gated output residual	42 only: a sign, not an effect size	the $+0.18$ and $+0.27$ of §3.1; §3.4
Decomposition, VT-2005-matched (`run_b_insuff_decomposition`)	none: the closure has no learnable parameters	§3.2.2; the estimator grids
Decomposition, broad IDAC (`run_b_insuff_representative`, `..._audit`)	none	§3.2.1: the $+0.51$, the $+0.95$, the deletion curve
The stratified map (`run_b_insuff_stratified_map`)	none	the 59 strata, Tables S8–S12
Closure-fidelity kernels (`run_fidelity_lever_fair2002`, `run_closure_flip_ci`)	none	§3.2.3, Table S6
Kernel residual (`run_local_closure_fix`)	five fit seeds; $\pm$ is their standard error	the $46 \pm 5\%$ and $-5 \pm 16\%$ of §3.2.4
Latent against its reference (`run_compensation_surrogate`)	$0, 1, 2$; mean$\pm$sd	§3.3, Fig. 4
Closure validation (`run_closure_reference_validation`, `run_published_idac_closure_check`)	none	the NIST reproduction; Table 2
External baselines (`run_external_baseline_comparison`)	one run each, on a *by-solute* split	Table 5
pKa second domain (`run_pka_real_decomposition`, `..._hammett_probe`, `..._binsuff_boot`)	K=20 within-scaffold splits; G=4 series as the cluster unit	§3.5
Noise floor and encoder probe (`run_noise_floor_estimate`)	none	§2.2, §2.1.2

One deposited artifact is defective, and the defect is disclosed rather than repaired: the seed-44 per-row oracle prediction file was truncated at save time to 295 rows. The three-seed summaries were aggregated at run time, before that file was written, so Fig. 2 and Table S2 are unaffected; any *per-row* re-analysis of that arm—Fig. 3 and the oracle row of Table 5—uses the two complete seeds and says so.

fiable where the VT-2005-matched set's 60 pairs are not, on two axes that are pure functions of molecular structure: an ordered nine-class cascade of substructure tests over the solvent, hydrogen-bonding role first and dominant functional group second, and the solute's role as water or as an organic. The classifier reads a SMILES string and nothing else—never m, g, the squared error, the source or the bootstrap—so no cell can have been chosen on its outcome, which is a property of the code rather than an assurance. A stratum is *boundable* at $n \geq 40$, where eight bins hold five rows; the smaller ones are reported at that same cell, with the resample-adaptive rule $\max(3, \lfloor n/10 \rfloor)$ carried beside them in the deposited table, nothing being dropped for being small.

The admissibility rule. A stratum is *admissible* only if it is *(a)* boundable at that fixed cell and *(b)* robust to leave-one-source-out—its margin keeps its sign when each source publication contributing to it is deleted in turn—in all four unit $\times$ convention cells, since the map reports all four. That last clause is load-bearing and is never dropped: without it the set taken whole passes too. A stratum drawing on one publication fails (b) by non-testability, there being no deletion to perform; a deletion leaving fewer than 16 rows leaves the fixed cell undefined, and fails (b) as well. Everything else is *reported in the map and not stated*, with its n, its sources, its margin, its interval and the reason. The rule governs the cells that favour us and those that do not alike, and it tests whether a number is stable, not what it means: an admissible stratum whose margin is not positive, and one that is another stratum restated, both establish nothing.

3 Results and discussion

Cluster-bootstrap frequencies P are quoted as bands—rounded to one decimal, with > 0.95 and < 0.2 closing the ends—because with 17 solvent and 41 solute clusters on the VT-2005-matched set they do not carry two-decimal resolution; the two-decimal values are in the deposited artifacts. Table 4 collects the load-bearing claims with the set each was measured on, its size, and the value with its uncertainty and scope. It is where the scope of every claim below is recorded, so that the prose states each finding once.

3.1 The two senses of grounding

The word "grounding" names two opposite operations on the same VT-2005 database. Supervising the σ head against the database *during training* improves solubility MAE from 2.043 ± 0.040 (ungrounded) to 1.846 ± 0.053 (grounded), a gain of 0.20 over three seeds. Substituting the reference profile for the learned one *at inference* degrades it, and that substitution alone is the effect this paper is named for. It erases any positive R^2 (Fig. 2): the model predicts solubility better *from its own learned σ-profile* than when *given the external reference one* (MAE 1.85 vs 2.25, in $\ln x_2$ units). Of the figure's eight arms two are non-COSMO controls, DirectGNN and a fitted NRTL closure; five COSMO-SAC arms differ only in how the σ head is trained and in what it is fed, and the sixth (*grounded+comb.*) differs instead in the closure, restoring the Staverman–Guggenheim combinatorial term. *Ref. σ (eval)* is the grounded checkpoint re-evaluated with the reference profile substituted at test time, *ref. σ (train)* the co-adaptation control injecting it during training instead.

The headline +0.41 is the evaluation-only substitution, and part of it may be unrelated to the closure: swapping an input distribution at test time degrades a pipeline on its own. The control that removes that component injects the reference profile *at training*, so the model co-adapts; at seed 42 it costs +0.18 ($1.803 \to 1.981$) where the evaluation-only substitution costs +0.43 at that same seed ($1.803 \to 2.232$), so part of the headline gap goes with co-adaptation and the co-adapted arm is by design the confound-free one. A channel-swap control costs +0.27, to 2.08. Both are worse than the learned-σ arm, which rules out an implementation artifact. Only the three-seed arms can be set against one another, and there the substitution costs about twice what supervision gains (+0.41 against −0.20); the co-adapted arm's +0.18 cannot enter that comparison.

Figure 3 sets the same rows out predicted against measured, for the control and three of the closure arms. Two things are visible there that the arm ordering is not. First, every arm is severely shrunk toward the corpus mean: the ordinary least-squares slope of predicted on measured is 0.515 for DirectGNN, 0.416 ungrounded, 0.528 grounded and 0.690 with the reference profile, none near 1, and in every panel the median trace crosses the identity—sparingly soluble compounds are called more soluble than they are and highly soluble ones less. That shrinkage is not the closure's doing: the closure-free control has the same slope as the grounded closure, so whatever produces it lives in the encoder and the data. Second, the reference-profile arm is the *least* shrunk and the worst. Its slope is the closest of the four to 1 and its prediction spread exceeds the measurement's, yet its MAE is the highest and its R^2 is not distinguishable from zero; substituting the reference restores spread that is not aligned with the truth, and its worst region is the sparingly-soluble tail it appears to reach. That is the misspecified map made visible on the solubility axis.

Which of the two channels the substitution fails through is not decided here. Drug-like solutes are essentially absent from the VT-2005 set, so the substitution falls almost entirely on the *solvent* channel, and the both-channel substitution the attribution would need covers only ≈7 solutes; enlarging that set would mean computing new reference profiles rather than looking them up. The causal claim is therefore carried on the activity axis, where the reference input is clean and matched on both channels: the decomposition (§3.2) and the latent's departure from its reference (n=44, §3.3). There the fixed reference-input closure sits well above the bound on what its inputs can fail to resolve, under-predicting $\ln \gamma_2^\infty$ with a systematic bias (Fig. S3).

Table 4: One row per load-bearing claim: what is claimed, the data set and its size, and the value measured with its uncertainty and its scope. Notes *a*–*h* qualify a row or a column.

Claim	Set (n)[a]	Value, with its uncertainty and scope[b]
The two senses of grounding (§3.1)		
Reference profiles hurt solubility at evaluation (MAE)	test split (5608), 3 seeds	$1.85 \to 2.25$; the per-seed values do not overlap
...and still hurt when injected at training	cross-arm rows[c], seed 42	$1.80 \to 1.98$ (own baseline $1.80 \to 2.23$); one seed, so a sign and not an effect size
...so part of that gap is test-time distribution shift	cross-arm rows[c], seed 42	share 0.59 on one seed: a description of that seed, not a magnitude
Training-time σ supervision *helps* (MAE)	cross-arm rows[c], 3 seeds	$2.04 \to 1.85$; the per-seed values do not overlap
Closure against inputs on the activity axis (§3.2)		
$B_{\mathrm{closure}} > B_{\mathrm{insuff}}$ on *glycol-ether* solvents[d,g]	broad IDAC, 182, 3 sources	$+2.04$ $[+1.27, +2.83]$ to $+2.94$, $P > 0.95$ in all four cells; worst deletion $+1.60$
... on every other stratum of the map	broad IDAC, 59 strata	§S3.1, Tables S8–S12, each with its margin, interval and the test it fails; none is stated as a finding[h]
... for either set *as a whole*	broad IDAC (477); VT-2005-matched (60)	$+0.51$ (full $+0.95$); -0.38 compounding two cuts ($+0.01$ full); $+0.18$ less one publication; $+0.05$ on the smaller set: an average weighted by composition[f]
$B^{\mathrm{up}}_{\mathrm{insuff}}$ near stratum-independent, MSE not	broad IDAC, 3 boundable classes; 6 with an estimate	$\times 2.0$ against $\times 25$; $\times 6.8$ against $\times 51$; p=0.02 against 2000 null partitions of 9 classes
$B_{\mathrm{closure}} > B_{\mathrm{insuff}}$, 2010 on its *own* typed latent[f]	broad IDAC (477); VT-2005-matched (60)	$+1.29$, $P > 0.95/{\approx}0.9/{\approx}0.9$[e]; -0.77 at $P \approx 0.3$ on the smaller set, where the ordering is lost: loss of resolution, not a reversal
Whether a better closure moves the ceiling (§3.2.3)		
The $2002 \to 2010$ step lowers the error	broad IDAC (477)	-0.99 at $P \approx 0.2$: no improvement, and not a measured equality either; under the full convention -0.07, $P \approx 0.6$, and $+0.06$, $P \approx 0.6$ with dispersion on
...but does on the VT-2005 set's chemistry	57 shared pairs, UD	$1.78 \to 0.77$, $P > 0.95$, at fixed profile database and temperature, full convention; those 57 pairs are also broad-set rows, so not an independent sample
A fitted kernel residual transfers across solvents	VT-2005-matched (60), grouped folds	$-5 \pm 16\%$: the interval spans zero, and excludes a cross-solvent reduction above about 30%
Mechanism (§3.3)		
The latent departs from its reference along a shared direction	matched molecules (44), 3 seeds	$51 \pm 2\%$; the magnitude is not stable across analysis configurations
...and that departure's top-two mode share exceeds a phase-randomised null	one grounded checkpoint, same 44	0.40 against 0.22 ± 0.01, the $\pm$ being the null's spread; one checkpoint, and a wrong-molecule null is not cleared
...and that direction *transfers* to molecules outside the fit	matched molecules (44), 3 seeds	$4.1 \pm 0.5\times$ a zero-drift null, with no constant-offset control on these seeds; the $\pm$ spans seeds at one partition
That departure *cancels* closure error	not measured	the one direct check is under-powered and runs against it
Accuracy (§3.6)		
Physics arm vs. DirectGNN accuracy (ΔMAE)	test split (5608)	$+0.14$ between separately tuned pipelines, so not attributable to the thermodynamic bottleneck

[a] Both sets are defined in Table 1. [b] P is a two-way (solute $\times$ solvent) cluster-bootstrap frequency that the margin stays positive, reported as a band by the rounding rule at the head of this section. [c] The cross-arm row set common to every arm of Table S2. [d] One estimator cell (§2.5) for both sets, both kernels and every stratum. [e] Bootstrap band over pair / two-way / source clusters. [f] An aggregate margin is a composition-weighted average, and the deletions and cuts recorded here are failures of the headline cell rather than robustness passes. These rows record what the aggregate was and how it fails; the rows above replace it. [g] The comparison is one-sided in both directions it can be read: a positive margin establishes $B_{\mathrm{closure}} > B_{\mathrm{insuff}}$, a non-positive one is failure to separate and licenses nothing. [h] Of fifty-nine strata, fifteen are boundable and six admissible, of which one row set is also positive and not contained in another—the glycol ethers, the one row set stated as a finding.

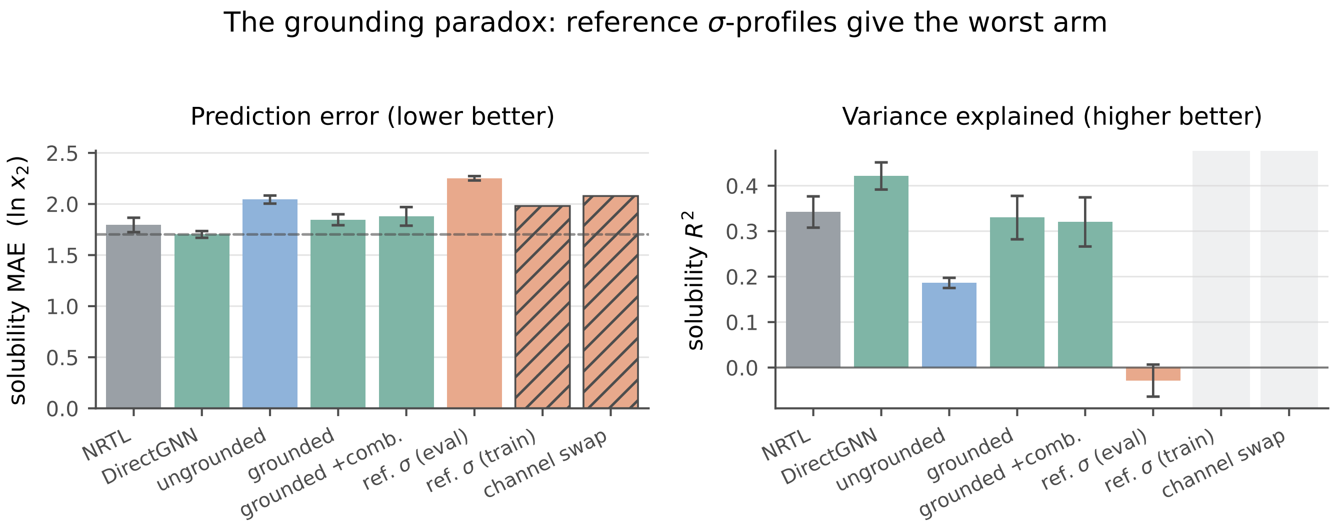

Figure 2: Controlled comparison on the scaffold split (n=5608). Solid bars: three seeds, mean ± a population standard deviation (`ddof`= 0). The two hatched bars are the co-adaptation controls, seed 42 only, so they carry no error bar and no R^2, and keep an empty slot in the right panel. Dashed in the left panel only, the DirectGNN reference.

3.2 Closure against inputs on the activity axis

The decomposition of §2.1.1 separates a fixed closure's error into a part in the map and a part in its inputs. Its output is a map over chemical strata rather than one verdict for a set, for the reason §3.2.1 establishes. It runs on two data sets that must not be conflated—the *broad IDAC set* (n=477, UD profiles, all temperatures) and the *VT-2005-matched set* (n=60 at 298 K, where the paradox's own database supplies both profiles)—and then asks whether a better closure moves the ceiling.

3.2.1 The broad IDAC set (n=477)

The closure reads a temperature as well as a profile pair and is evaluated at each row's own temperature, so the conditioning variable is its full argument $(z^\star, T)$—the concatenated UD profile pair with T, 103 dimensions against n=477 (dim $/n \approx 0.22$)—and the target m is the experimental $\ln \gamma_2^\infty$, so floor and closure error are measured on one data set and one profile database. Its chemistry is small: no solute exceeds 12 heavy atoms (median 7), against a median of 14 and a maximum of 122 for the drug-like solids of this paper's own corpus, and only 0.46% of corpus rows carry a solute from this set. *Broad* therefore names the reach of the γ^∞ corpus the kernel is calibrated on and not coverage of the chemistry a solubility model is deployed against: it covers the solvent side of deployment and not the solute side. It is matched to UD by InChIKey with a first-block fallback, looser than the VT-2005 set's rule; a wrong $z^\star$ inflates B_{closure}, so the looser rule runs against the ordering we claim, and the aggregate margin is unchanged on the 465 exactly matched rows.

Under the convention rule of §2.5 the fair 2002 arm scores MSE $= 1.902$ against $\text{Var}(m) = 4.85$, the LOTV bound gives $B_{\text{insuff}} \lesssim 0.70$ and hence $B_{\text{closure}} \gtrsim 1.21$: a separation margin of +0.51 for the set taken whole, +0.95 under the full convention and +0.27 collapsed to one row per distinct pair, positive in every cell of the bin-count × variance × convention grid, with two of the three bootstrap clusterings extending just below zero.

That aggregate is not the instrument's output. Two operations reported separately—the pair unit, and dropping the two chemistries the error concentrates in, the glycol-ether solvents and the rows carrying water as the solute—compound badly. The chemistry cut alone takes +0.51 to −0.12 on the 258 rows that re-

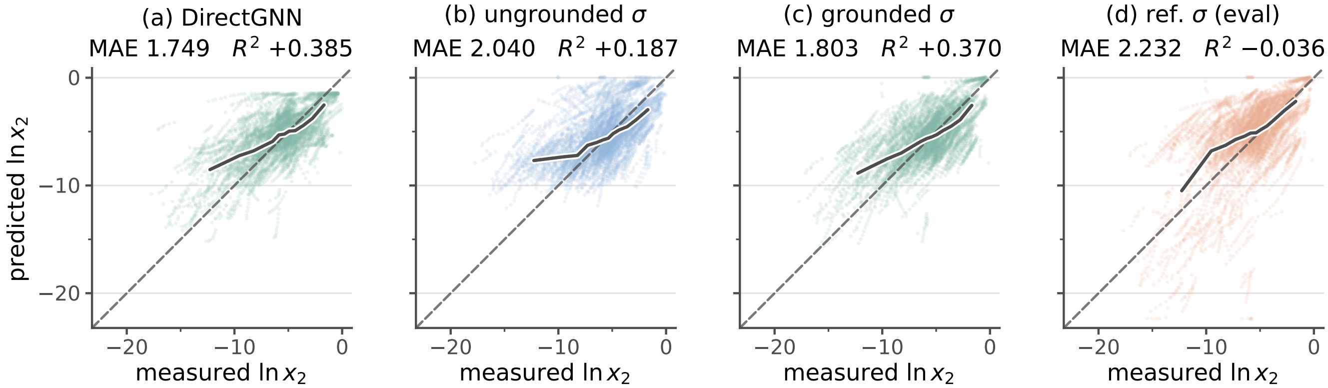

Figure 3: Solubility parity on the scaffold-split test rows: predicted against measured $\ln x_2$ for four arms of the controlled comparison. All four panels are the same 5608 rows—the cross-arm lock, rows supervised and finite in every arm—from seed 42 of three. Those rows are 763 solute–solvent pairs over 147 solutes and 70 solvents, measured between 248 and 368 K, so one pair contributes a short temperature series and the number of points exceeds the number of independent systems. (a) is the closure-free control; (b) is COSMO-SAC over an unsupervised σ-profile, (c) the same closure with σ-profile supervision during training, and (d) the arm of (c) with the reference VT-2005 profile substituted at evaluation. Dashed, the identity; solid, the median prediction in twelve equal-count bins of the measurement. The MAE and R^2 above each panel are that panel's own, on those rows and that seed, to the three decimals Table S2 uses for a per-seed value; the three-seed means are Fig. 2 and that table, and differ from these by at most 0.05 in MAE. In every panel the median trace crosses the identity: predictions are too soluble where the measurement is small and too insoluble where it is large. Only (d) carries scatter below the identity that reaches $\ln x_2$ values no measurement occupies. The axes span the union of measured and predicted values over all four arms; no point is clipped.

main, the pair unit alone gives +0.27, and the two together give −0.38 on 136 pairs; under the full convention the same pair of cuts takes +0.95 to +0.28 to +0.01. Nor is the aggregate carried by the set: deleting one source publication at a time, `10.1021/acs.jced.7b00114`—37 rows, six molecule pairs, 32% of the set's squared error—takes the margin to +0.18, while each of the other fourteen deletions leaves it between +0.42 and +0.87 (under the full convention that curve is flat, [+0.82, +1.07] against +0.95). That single deletion loses two thirds of the margin and its resolution without reversing its sign; the sign reverses only under the compounded cut. Two caveats qualify the corroborating estimators and not the bound: under random folds they leak, and under folds blocked on molecules they are uninformative on the VT-2005-matched set and split on the broad set; and the binning-free pair bound, tighter though it is, is not promoted, because its multi-row pairs each come from a *single* publication and it therefore samples no between-laboratory component. Together they say the binning bound is slack rather than that B_{insuff} is negligible. What is reported rests on the LOTV bound, which fits no model and so has no fold design to leak through (Table S7).

The map, and how much of it survives the rule. The instrument runs over 59 strata of the broad IDAC set—477 rows, 185 pairs, 78 solutes, 36 solvents, 15 source publications—on the solvent axis at three granularities and the solute axis at two, plus their cross-tab and the set as a whole. Fifteen have $n \geq 40$ and are boundable, the only ones whose margins the rule can act on; the other 44, including the 24 that carry no estimate at all, are printed with the test each fails in §S3.1 (Tables S8–S12 and Fig. S2), every stratum with its error, its bound, its interval and its share of the set's squared error, and nothing dropped for being small. Six of the fifteen are *admissible*. Those six are three distinct row sets, of which one is both positive and not contained in another: the glycol-ether solvents. **That row set is the one established finding**: on 182 rows over 43 pairs from three publications carrying 45% of the set's squared error, the closure's own error exceeds what input insufficiency can account for, by +2.04 with a 90% interval of $[+1.27, +2.83]$ at the deployed convention—+2.94 at the pair unit under the full one—positive in all four cells and under every single-publication deletion, the worst leaving +1.60 when `10.1016/j.jct.2016.10.013`, which carries 53% of that stratum's squared error, is removed. It is not an artifact of one solute family: the 74 rows left when its alkane solutes are dropped keep the sign at +1.40 deployed and +1.69 full, under all three deletions.

Which chemistry does not separate. Three of the six admissible cells are that same row set at three granularities. The remaining three are two row sets, and neither establishes anything. The alkane solutes (129 rows, 45 pairs, +1.38, $[+0.79, +2.33]$) are 84% the glycol-ether rows and return -0.22 on the 21 outside them—itself a sub-threshold number, so the demotion rests on the overlap arithmetic and not on it. The acceptor-only aprotics (57 rows, -0.19, negative in all four cells, $P < 0.2$ at both units under the deployed convention and $P \approx 0.3$ under the full one, at an MSE of 0.089 that is at the scale of the measurements) pass the rule and license nothing, the instrument being one-sided; their deletion test is additionally *anti*-conservative, two of the four deletions leaving $n < 40$, where the Bessel within-bin variance inflates—0.140 to 0.258 on the n=38 remainder—and drives a negative margin further negative. That is the whole of the uniqueness claimed for the glycol ethers: the only row set admissible, positive and not contained in another—*not* the only admissible stratum and not the only one sign-stable under every deletion. Of the rest, water carried as the dilute solute (37 rows, +6.33) and the halogenated solvents (30 rows, +3.39) are too small to bound and draw on one publication whose deletion cannot be run, so those two large margins were never put at risk; water as a solvent, the coarse hydroxylic class and the organic-solute subset each lose their sign under a deletion, which is why the finding belongs to the glycol-ether sub-class and not to the hydroxylic solvents as a class; the mono-alcohols, the N–H protics and the ester and azine subfamilies, which an acceptor-only reading would lean on, are unboundable with a deletion that cannot be run. None may be used to explain a reversal. The whole-set row passes at the headline cell and fails once the pair unit is required, so the rule does not by itself dispose of the aggregate there; the composition argument above does.

Why the output is a map. $B^{\mathrm{up}}_{\mathrm{insuff}}$ is nearly stratum-independent and MSE is not. Across the three classes the design can bound—glycol ether, water and the acceptor-only aprotics, bounds 0.108, 0.215 and 0.140—the bound spans a factor of 2.0 against errors (2.252, 0.476, 0.089) spanning a factor of 25; adding the three that carry an estimate without being boundable—mono-alcohol 0.733, halogenated 0.592, N–H protic 0.281—gives 6.8 against 51, that 6.8 being set by the mono-alcohols' 0.733, which exceeds that stratum's own $\mathrm{Var}(m) = 0.604$ and is flagged in the artifact as an invalid bound on a stratum already excluded. The two orderings are unrelated—the glycol ethers carry the lowest bound of any class and the second-highest error—and against 2000 random partitions into blocks of the observed class sizes, at the same cell, the chemical partition's spread in MSE exceeds every draw while its spread in the bound is unremarkable ($p = 0.26$); their ratio gives $p = 0.02$. That is the estimator showing through: every stratum is summarised by eight quantile bins of one scalar, so the bound measures the residual spread of m inside such a bin, a property of the binning rather than of the chemistry, while nothing constrains the error to follow it. An aggregate margin is therefore a composition-weighted average of a term free to move 25-fold offset against one that barely moves, and it will report a binding closure for any set whose composition happens to include a chemistry the closure fails on—a statement about the aggregate, not about the bound, which stays valid in each stratum where it is computed.

3.2.2 The VT-2005-matched set (n=60)

This set carries the paper's other reference-matched analyses, and it is far weaker. All 60 rows sit at 298 K, so $z^\star$ is the profile pair alone and $\dim(z^\star)/n \approx 102/60$ leaves the split behind one-sided bounds; it is strongly unbalanced, methanol occupying 25 of the 60 rows and the two worst-fit pairs carrying 34% of the squared error; and its 17 solvents fall in six of the nine classes, none reaching the n=40 a bound needs, so it supports no stratified bound and no stratum of it is stated as a finding. Its aggregate margin is +0.05, and it does not stand: deleting the two leverage pairs takes it to −0.002, dropping methanol loses it, blocked out-of-fold estimators do not support it, and 11 of the 60 single-pair deletions take it non-positive, where on the broad set no single-pair deletion moves the sign at all. Zero-mean label noise can understate a separation but not create it (§2.1.1); what a margin does bound is a label systematic tracking $z^\star$, implausible on the maximum-likelihood binning and *not excluded* on the Bessel-corrected one. Of the 60 pairs, 57 also appear in the broad IDAC set at 298 K under UD profiles, so the two are not independent samples—this is the sub-design VT-2005, and therefore the paradox's own oracle, is defined on, and the agreement between the two sets' error orderings is a consistency check rather than corroboration.

3.2.3 Whether a better closure moves the ceiling

If the ceiling is the closure, the direct remedy is a better closure. The standard 2002 → 2010 step delivers one on the VT-2005-matched set and not on the broad IDAC set, and a block split identifies the factor that carries that reversal.

The two kernels have no *prescribed* common input, σ-profile averaging being part of each model's definition, so evaluating each on its native profile is the specified use. An input-fixed *control*—profile database, cavity area, volume and averaging held fixed, kernel alone varied—gives 2.047 → 0.765 on the smaller set ($P > 0.95$), under three limits: it cannot separate the donor/acceptor typing, which the 2002 kernel has no channel to consume and which is therefore charged to the kernel by construction; its step is larger than the native-input one, so it is favourable to the conclusion it supports; and its re-averaging term is unresolved. The NIST implementation detail fixing the *fair 2002* arm at 1.757, and the three arms on both sets under both conventions, are in Table S6.

The step more than halves the smaller set's error under the full convention ($P > 0.95$), the convention under which the two kernels' published error figures are reported, and cuts it 42% under the deployed one. On the broad IDAC set it is not an improvement under either rule, and the 3% the dispersion term recovers there is a null whose interval is an order of magnitude wider than the effect. That arm's run on the smaller set is on n=57, three halogenated pairs having no tabulated dispersion value—a coverage limit of the model offered as the lever; no claim rests on the 0.765-versus-0.785 ordering, which is not an artifact of the differing n.

The reversal is the *chemistry* and neither the profile database nor the temperature range: with UD profiles, one extraction and one convention throughout, the *2010, dispersion off* arm wins on the 57 pairs shared with the smaller set, does not win on the complementary 420, and still does not win on the 80 complementary rows that are *also* at 298 K—that last block at $P \approx 0.3$ and no interval, the three blocks not having been tested against one another. The VT-2005-matched set is a low-γ, hydrogen-bond-dominated sub-design, the regime the donor/acceptor-typed 2010 kernel was built for. The map does not corroborate the reversal from the other side and we do not ask it to: the typed kernel is not an improvement on the broad set, of whose squared error the glycol ethers carry 45%, and the classes that would explain a reversal—the halogenated solvents, the aryl ethers and water as the solute, largely one row set, all 13 aryl-ether rows and 18 of the 30 halogenated being water-as-solute rows from one publication—are the cells the rule excludes. The block split is the whole of the evidence for the reversal being chemical.

The floor checks are same-database, same-latent *and* same-convention. The *UD-native* check on the smaller set, the one commensurable with the broad set, is at zero under the deployed rule and +0.69 under the full convention. Under the 2010 kernel's own typed latent, 306-dimensional and so several times worse resolved, the ordering is *lost* on the smaller set (−0.77) and holds on the broad set (+1.29)—both aggregates, and read as such. That is loss of resolution, not a demonstrated flip to the inputs, which would need a *lower* bound on B_{insuff}. We know of no large-scale infinite-dilution benchmark of the 2002 → 2010 step, and our search was not exhaustive, so the broad-set result is an absence of evidence rather than evidence of absence; it lies within the range of magnitudes the comprehensive assessment[29] reports. We claim only that **the fidelity lever we can demonstrate is confined to the smaller set's chemistry**, and that null is additionally a small-molecule and glycol-heavy one, so on drug-like solids the lever is untested rather than absent.

3.2.4 Correctability, and where the closure fails chemically

Where B_{closure} dominates—on this set, the one row set the rule establishes—the ceiling is in the map; that does not say the map is *fixably* wrong. A small structure-preserving residual on the fixed COSMO-SAC exchange-energy kernel, fit on the reference VT-2005 profiles, removes $46 \pm 5\%$ of the reference-input error under random folds but $-5 \pm 16\%$ once folds are grouped so that no solvent is shared. Only the grouped design is free of leakage, the closure error being predominantly a between-solvent systematic. Stated as a bound rather than a null, that design excludes a cross-solvent reduction above about 30%. B_{closure} is therefore a defect a handful of kernel parameters can absorb *within* a solvent and that we have not shown to be correctable across solvents.

Where it fails is chemically placed, and placed differently on the two sets. On the VT-2005-matched set 90% of the squared error falls on strong hydrogen-bond-donor solvents and the closure is near-exact on acceptor-only ones, putting that set's B_{closure} in COSMO-SAC's documented single-parameter H-bond kernel—the specific defect the donor/acceptor-typed 2010 closure was built to fix, and the same localisation the published evaluation of Table 2 shows on 130 times as many rows. On the broad set that reading does not reproduce, and neither does its replacement, an absent dispersion term: which of the two the broad set points at is decided in either direction by the same publication the map already declines to state, so we state neither, and the H-bond reading remains the VT-2005-matched set's, neither corroborated nor replaced by the broad set. The residual's fold design and rank ordering, its failure to gain over the free 2002 → 2010 step above, and the Staverman–Guggenheim counter-test in which the physically *more* complete combinatorial term *increases* error on size-asymmetric pairs are in §S3.

That COSMO-SAC-2002 is a low-fidelity closure is the contribution rather than a qualification of the result: it is what a practitioner adopts by default, a closure's fidelity is *not knowable in advance* on a new system, and the field routinely composes an expressive encoder with a fixed mechanistic map and assumes the map is good enough. What follows when it is not is that the learned latent silently departs from the quantity it is named after (§3.3), invisibly without an external oracle. Outside that one row set the map reports cells it cannot resolve, not cells in which the closure is sound. One qualification attaches to B_{closure} itself: the ground truth charging it is quantum chemistry, not experiment—the VT-2005 profiles are computed for a single reference conformer and protonation state, and the successor Delaware database revises those conformations for about a third of its compounds.[29] If the reference is displaced, even a perfectly specified g carries a term in B_{closure}, whose share we do not bound, so what is measured is the misspecification of a composite object, the kernel applied to one fixed computed conformer. The deployed pipeline is that same composite, so this is the quantity a practitioner faces.

3.3 The end-to-end latent against its reference

A candidate *mechanism* for the low-fidelity, end-to-end regime is this: a σ-profile head trained *end-to-end* through a fixed, misspecified closure—with no σ supervision in that phase—is under pressure to *cancel* the closure's error rather than to be physically faithful. The hypothesis has two parts and this section establishes only the first. Its antecedent—that the latent leaves the reference along a direction shared across molecules rather than as per-molecule noise—is measured. Its consequent—that the departure cancels part of B_{closure}—is not, and the one direct check available argues against it. The external reference that fixes the ceiling makes the first part testable: we compare the predicted profile $\hat{\sigma}$ against the reference VT-2005 profile molecule by molecule (n=44).

The drift under fine-tuning. The comparison is within a single grounded model, checkpointed before and after solubility fine-tuning (the σ head unfrozen), so architecture and initialisation are held fixed and only the objective differs; it is repeated over *three seeds* and reported mean±sd. Under σ supervision the head reproduces the reference profile to within 36±1%, so the intermediate *can* carry the physical shape, though only as an in-sample fit: all 44 probe molecules lie inside the σ-grounding pool. Solubility fine-tuning then drives the profile $41 \pm 3\%$ off the grounded one, for a total departure from the reference of $51 \pm 2\%$. The drift is not per-molecule noise, and its structure is not a constant offset: the first two principal components of the *mean-centred* deviation carry $72 \pm 1\%$ of the variance. The quantity read against a null is a different one, the top-two share of the learned-versus-reference deviation, 0.399; its null is *phase-randomised*, matching each deviation's own smoothness and destroying only the between-molecule alignment at issue, and the observed share clears it (0.399 against 0.221, $p < 0.001$), at roughly 1.8×. The 72±1% mean-centred share has no matched null of its own, a wrong-molecule null built from real reference profiles does *not* clear (0.772, $p = 1.00$), and if the case for preferring the phase-randomised null is not accepted the spectrum is uninformative rather than supporting. To test whether the structure transfers across molecules we fit one distortion operator D on half the 44 molecules and score it on the other half, where it predicts the drift 4.1±0.5× better than the zero-drift (identity) null out of sample. That null is deliberately weak, nothing controls it on these three seeds, whose checkpoints were not retained, and a molecule-independent constant offset reaches 2.1× on the one companion run where that control can be computed, so the ratio stands against the identity null alone; the held-out half is homologue-saturated. Structure *beyond* an offset therefore rests on the mean-centred spectrum, not on the ratio—and that spectrum does not exclude a shrinkage of every profile toward the corpus-mean shape. On the reference ladder of §S5 the drift consumes essentially all the molecular specificity grounding had provided, as such a collapse would also look.

Dominant mode, and the direct cancellation test. The dominant drift mode (Fig. 4) transfers area *out of* the nonpolar bins ($\sigma \approx 0$) and *into* the polar wings: the network represents molecules as more polar than they are. Whether that shift *cancels* part of B_{closure}—the input compensating the map, as the non-identifiability of the split allows (Lemma S1)—is the comparison $\mathrm{MSE}(m, g(\hat{\sigma}))$ against $\mathrm{MSE}(m, g(z^{\star}))$ on the same pairs. It does not support the cancellation reading. Holding the cavity area at its reference value so that only the profile shape varies, the sole retained end-to-end SLE-trained COSMO-SAC checkpoint gives 18.75 against 1.47: on the activity axis its drifted profile is an order of magnitude *worse* through the same closure, not better. Three limits sit on that number. The checkpoint is the uncontrolled one, whose departure (82%) is half again the three-seed $51 \pm 2\%$. The axis is wrong: the drift is produced

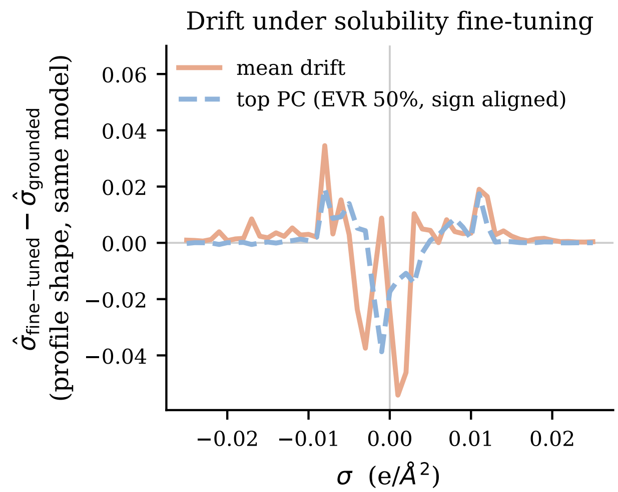

Figure 4: The drift solubility fine-tuning imposes on the learned σ-profile (n=44 VT-2005-matched molecules): $\hat{\sigma}$ after fine-tuning minus the *same model's grounded profile*, not minus the external reference. Shown are the mean drift and its dominant PCA mode, defined only up to sign and drawn in the mean drift's direction; the legend's EVR is that mode's explained variance ratio.

by the finite-composition solubility objective and scored here at infinite dilution. And the check has almost no power in the direction that matters, so it separates "no cancellation" from "large cancellation" but not from "some cancellation". It is the only direct evidence available, and it is evidence against rather than a clean pass or fail. A second, independent check on the closure's own axis points the same way: a *more* polar profile moves g further from m (§S5). Both sit on that same infinite-dilution axis, so neither refutes cancellation in the regime that generates the drift; they remove the one place it could have been measured, and the cancellation reading remains inferred from the paradox itself. What is established is weaker and still consequential: the model's "physical" intermediate is not the molecule's charge distribution, consistent with cautions that the interpretability of physics-grounded latents is not guaranteed under misspecification,[20,30] and that adapting a representation can itself collapse an otherwise physically-faithful latent.[31] Its *structure*—low-rank rather than diffuse—excludes a random collapse but not the structured shrinkage above.

What leaves σ under-determined by activity. The freedom that would make such a cancellation possible is the closure's geometry: at infinite dilution the activity map concentrates on the one or two hydrogen-bonding directions where the closure is least faithful. The end-to-end head can therefore move σ along the remaining, low-sensitivity directions at little cost to the fit—the σ-profile form of Lemma S1. The Fisher measurement behind that statement, and the two respects in which it does not settle the question, are in §S5; nothing here shows recovery of the physical profile to be impossible in principle. The sign of grounding turns on *both* axes, and on which sense of grounding is meant. Freezing the σ head at its supervised warm-up state instead of letting it drift end to end cuts the reference-input penalty by 65% (ΔMAE from +0.40 to +0.14 over n=5571 matched test rows). Those are a separate matched pair of runs, one base resumed twice with the freeze off and on, and not the +0.41 headline arms of §3.1. A near-physical intermediate makes the pipeline far more robust to substituting the reference σ. Supervision alone does not, however, *flip* the sign here—the penalty stays positive, the closure still being the low-fidelity COSMO-SAC-2002 and the warm-up σ itself only partly physical. The full flip needs supervision *and* fidelity together, which is what TeNNet-SAC[7] has and this pipeline does not.

3.4 A downstream correction on the output

If the closure binding were merely an inaccurate point estimate, a learned output-space correction on top of the physics prediction should recover accuracy. It does not: an additive bounded residual with a learned confidence gate leaves the gate effectively shut, and forcing it open removes the systematic *bias* (+0.45 → +0.06) but recovers little *accuracy* (R^2 0.236 → 0.274 on the same rows), as expected if the closure, not the estimate, is the binding constraint.

Both arms are seed 42 only, and this is a statement about the one correction tested: an additive bounded residual on the output, gated, on this closure and this corpus.

3.5 A second chemical closure: pKa through a fixed Hammett relation

The instrument is not specific to COSMO-SAC, and a second domain runs it where the sign of grounding can be predicted before the measurement. The closure is a Hammett linear free-energy relationship $g(\sigma_\mathrm{H}) = \mathrm{p}K_{a,0} - \rho\,\sigma_\mathrm{H}$ in a substituent constant; the learned intermediate is that constant predicted from structure; the external reference is the Hansch–Leo table; and the data are the OPERA experimental compilation[32] under a uniform, closure-independent quality filter that drops pre-ionized microstates. The two classic failure modes of that relation fall cleanly onto the two channels of Lemma 1: *resonance saturation*, a curvature the linear closure cannot represent, lands in B_closure, while *ortho/steric* effects, absent from the tabulated constants, are information the intermediate lacks and land in B_insuff. The two poles differ accordingly. On the meta- and para-substituted aromatics the relation was built for (n=158) it is well specified: RMSE 0.55, with B_closure below the insufficiency floor. On the ortho-substituted and heteroaromatic ones it does not describe (n=846) it degrades to RMSE 2.26, of which a within-scaffold out-of-fold estimator places at most 4.5 of the 5.1 total in input insufficiency and hence at least 0.6 in the closure—both figures one-sided, in the same direction as the solubility bounds.

The decisive test is the same oracle swap, measured *in-distribution* across K=20 within-scaffold splits: the learned $\hat\sigma_\mathrm{H}$ replaced by the tabulated $\sigma^\star_\mathrm{H}$ in the same closure, the sign of the change recorded per stratum. It flips. On the clean pole the reference *improves* accuracy—MAE 0.48 learned against 0.315 with the reference, better in 19 of 20 splits and in three of the four reaction series—so the ceiling there is the noisy learned estimate and not the closure, which is an internal positive control on real data. On the ortho pole the same swap *degrades* it—0.83 against 1.62, worse in 20 of 20 splits and in all four series, gap 90% CI $[+0.57, +0.95]$—the freely learned constant absorbing the ortho and steric information the tabulated one lacks. The sign of grounding therefore reverses inside one closure, in the direction Proposition 1 gives a priori and on the channel it names: input insufficiency, rather than the closure-error cancellation of §3.3. Both signs obey one crossover law—grounding helps a pole iff the learned arm's error there exceeds the fidelity-set oracle error—which a competence sweep confirms: the clean pole's threshold ($\approx$ 0.315) is never beaten by the learned arm, so grounding helps at every training budget, while the degraded pole's threshold (1.62) is high enough that the learned arm beats it across the whole interpolation range, down to 12% of the data, and grounding reverts to helping only under scaffold *extrapolation*, where the learned constant breaks (2.6±2.0) back above the threshold. The clustering unit is the Hammett reaction series, of which only G=4 exist, so the bootstrap resolves nothing finer than series-level sign agreement, read as a sign test on four units (p=1/16 on the pole where grounding hurts). The construction, the estimator validation and the measured fidelity sweep are in §S7.

3.6 Scope of the results

The result is a cautionary measurement about one of the two operations the word "grounding" names. Supervising a learned physical latent on reference values during training helps; *supplying* those values at inference hurts, and it hurts because the fixed map consuming them is misspecified. When that map is of unknown fidelity—and the standard route does not improve it—better inputs surface its bias rather than help. Both sets are scoped by the reference database that defines them and neither is the deployment regime, so the fidelity null is a null on small-molecule chemistry and the lever remains untested on drug-like solids. The useful form of the finding is one-sided and narrow:

a place where grounding effort is *not* the remedy, on one row set of the map, rather than an allocation across chemistries.

The unconstrained control's representation. If the physics arm's intermediate is a surrogate, the thermodynamics might still be present implicitly in the control, which is free to represent it. It is not. A ridge probe of the control's pair representation recovers the model's own prediction at $R^2 = 0.98$ but not the measured ideal-solubility term—an upper bound at $R^2 \lesssim 0.30$ rather than an exclusion (limitation (ix)). What the representation is organised by instead is scaffold identity, consistent with the transfer limitation of §2.1.2: a test molecule on an unseen scaffold is not a small extrapolation but a point off the axis the representation is organised along. The seed-by-seed scores, the permutation nulls, the within-scaffold measure and its weakness, and the temperature probe are in §S10.

The contribution. The unit of contribution is the measurement, not any single number it returned. Its two ingredients are individually standard—the closure–insufficiency split is a law-of-total-variance identity, and the sign of grounding's effect follows the model-discrepancy principle[19,20]—and what is new is their combination into a usable instrument: an identity on a *fixed* operator, with the insufficiency side bounded rather than point-identified, whose sign rule can be tested *a priori* and *bidirectionally.* The pKa domain supplies that test as an in-distribution sign flip, at the resolution four series allow and no better—an oracle-on-a-fixed-operator affordance the concept-leakage setting lacks, its intermediate having no ground truth.[17,18] The solubility results are the instrument's first application, and their transfer out of the matched regime is not established.

Where the accuracy stands, and the confound in it. Table 5 places every arm of this work beside the external models, in three blocks that may not be read across. The physics arms trail their own control, and the predict-the-mean row shows how little room sits above them. The physics and control arms were tuned separately and no hyperparameter-matched control was run, so the global ΔMAE of +0.14 between them carries no information about the thermodynamic bottleneck, and neither do the per-solvent-class map, the per-solute-class map, the novelty inversion or the data-efficiency trend, all of which resolve the *same* difference along chemical axes. We claim no accuracy result in either direction; those strata are reported in §S6.4 and §S6.5 as descriptions of where two particular pipelines differ, and no conclusion rests on them. Running the matched control is the direct repair and we have not run it. What survives the confound is every *within-checkpoint* contrast, where the two sides share one trained model and one tuning: §3.1, §3.2 (which uses no trained model at all), §3.3 and §3.4. Separately, and not as an accuracy claim about the bottleneck: we did not find a configuration in which the physics path attains a lower error than the control, and grounding the crystal factor on external labels did not move the ceiling (§S6.3). The one place the physical *structure* helps is temperature extrapolation, as a hypothesis-class property the trained models do not exploit (§S6.1).

The external block is not like-for-like and the table says so in its own rows. FastSolv and SolProp are deposited here on a *by-solute* split, this work's arms on the harder solute-scaffold split; retraining them on the scaffold split needs a GPU run we did not perform, and scoring released FastSolv weights on our test rows is invalid because FastSolv is trained on BigSolDB, which is this corpus. The third block is as published, on other corpora, with $\log_{10} S$ in molarity against mole fraction here, so the unit matches and the task does not. One sentence survives all three caveats: this work's control, at $\log_{10} S$ RMSE 1.00 on held-out scaffolds, is inside the band published new-solute extrapolation occupies—0.83 to 0.95 for the leaders, 1.43 to 2.16 for the physics-informed cycle—rather than an order away from it, and no arm in any block is near the 0.75 aleatoric estimate,[1] itself a different functional on a different base from this paper's own floor (§2.2). How much the split matters is visible inside the table: the same FastSolv retraining scores 1.94 by solute and 2.91 on random pairs.

The sign rule's own prediction. Proposition 1 predicts *when* to expect these negatives: a prior hurts precisely in the mature-data, low-noise, well-covered regime. The synthetic sweep (Fig. S8) is true by construction there, so it verifies the instrument rather than the hypothesis, while the pKa closure exercises the sufficiency channel on real data. Whether grounding helps is shaped by *two* axes, closure fidelity and how the latent is trained; characterising that plane on real systems is future work. The chemical strata form a pattern in the direction the sign rule would give, but separately tuned pipelines can produce it for reasons unrelated to the prior and there is no multiplicity control, so §S6.4 records it as the experiment a matched control should run and the diagnostic is *proposed* rather than validated.

Limitations: what the sets can carry. (i) The phenomenon and its causal attribution lie in *different regimes*: the paradox is measured where the model operates (finite composition, all γ, n=5608), the attribution and the latent-departure result only on infinite-dilution activity data (n=477 across temperatures, and the 298 K, low-γ, VT-2005-matched n=60 and n=44 sets). Transfer back to the paradox's own regime is a conjecture and is the principal open question here. The n=60 set is narrowly sourced—all by gas chromatography and all from three publications, 33 pairs from one—so neither a method-stratified nor a source-clustered re-estimate is available on it, and its margin is extraction-sensitive: a broader ThermoML pull at 298 K widens it but *inverts* it on the gas-chromatography subset, a sensitivity we leave unresolved. The broad set does not move that way under the same cut and is not the set the paradox's own oracle is defined on; the arithmetic and the enlarged-pull values are in §S9. (ii) Under the deployed convention no *aggregate* margin is resolved on either set and only the literature-standard convention resolves one, while the headline convention is the deployed one; the aggregate falls for the stronger reason of §3.2.1 rather than for want of resolution. (iii) Of the map's six admissible cells the glycol ethers are the only row set also positive and not contained in another, and not the only cell whose deletions all run; the halogenated solvents and the mono-alcohols keep their sign under the one deletion each admits, so they are unverified rather than shown to fail.

Limitations: what the estimator can deliver. (iv) B_{insuff} is upper-bounded, not point-identified (no replicates at fixed $z^{\star}$); the smoothness-free Jensen bound assumes m and g share the $\ln \gamma_2^{\infty}$ reference state, established by the closure validation of §2.3, so the mean offset is decoder bias and not a units artifact. (v) The constant-offset margin is convention-specific and inverts under the deployed default on both sets; there the separation rests on the law-of-total-variance bound and the more-physics-worse effect. That bound is not itself convention-independent—only the estimand B_{insuff} is, Lemma 1(d)—so the deployed claim rests on it *computed under the deployed convention*. (vi) The stratification reads structure alone, so no cell was selected on its outcome, and it is exhaustive and disjoint *within* an axis but not across the two axes or their granularities: 362 of the 1711 stratum pairs share rows, all 37 water-as-solute rows lie inside three solvent classes, and 108 of the 129 alkane-solute rows are glycol-ether rows, so cells from different axes are not independent evidence and those rows are counted once. Granularity is an analyst choice, which is why three solvent levels and two solute levels are all reported. There is no multiplicity control on the per-cell bootstrap frequencies, so one cell's P describes that cell alone; the admissibility rule is the only protection in place, six of the nine solvent classes are not boundable here and three carry no estimate at all. The rule is also one-sided in what it can deliver: with B_{insuff} bounded only from above, no cell can ever return "better inputs would help here", so the map only ever withholds a chemistry from an allocation of grounding effort.

Table 5: Solubility accuracy in three blocks that may not be read across: this work, external models re-run here on a different split, and published figures on other corpora.

Model	Test set	n	MAE $\ln x_2$	RMSE $\log_{10} S$	R^2 $\ln x_2$
This work, solute-scaffold split; three seeds, mean ± s.d.					
DirectGNN, same encoder, no closure *(control)*	solute scaffold, this corpus	5608	1.70 ± 0.03	1.00	0.42
TGNN, NRTL closure (physics)	solute scaffold, this corpus	5608	1.79 ± 0.07	1.07	0.34
TGNN, COSMO-SAC closure, supervised $\hat{\sigma}$ (physics)	solute scaffold, this corpus	5608	1.85 ± 0.05	1.08	0.33
TGNN, COSMO-SAC, reference $\sigma^\star$ substituted at evaluation[a]	solute scaffold, this corpus	5608	2.26 ± 0.02	1.35	−0.05
predict the test-set mean of $\ln x_2$	solute scaffold, this corpus	5608	2.32	–	0.00
External models re-run on this corpus—a different split, so not a ranking against the block above					
FastSolv, retrained on this corpus	by-solute, this corpus	10287	1.94	1.10	0.23
SolProp cycle, released model + 3-parameter recalibration	by-solute, this corpus	10287	2.34	1.34	−0.18
SolProp cycle, released model, zero-shot	by-solute, this corpus	10287	2.36	1.87	−0.47
SolProp encoder retrained on $\ln x_2$, cycle discarded	by-solute, this corpus	10570	1.62	1.07	0.39
FastSolv, retrained on this corpus	random pair, this corpus	10696	2.91	1.54	−0.59
SolProp encoder retrained on $\ln x_2$, cycle discarded	random pair, this corpus	10887	0.64	0.45	0.88
As published, on their own corpora and splits; $\log_{10} S$ *in* $mol\,L^{-1}$					
fastsolv (solution-FASTPROP ensemble)[1]	SolProp test set	–	–	0.83	–
solution-CHEMPROP (companion)[1]	SolProp test set	–	–	0.83	–
SolProp cycle,[2] scored in[1]	SolProp test set	–	–	1.43	–
fastsolv (solution-FASTPROP ensemble)[1]	Leeds set, near room temperature	–	–	0.95	–
solution-CHEMPROP (companion)[1]	Leeds set	–	–	0.99	–
SolProp cycle,[2] scored in[1]	Leeds set	–	–	2.16	–

[a] Seeds 42 and 43 only: the seed-44 per-row file for this arm is the truncated deposit of Table 3. The three-seed bar of Fig. 2 for the same arm, 2.25 ± 0.02, was aggregated at run time and is unaffected. $\log_{10} S$ is converted from $\ln x_2$ through the same converter for every row of the first two blocks; the third block is quoted as published, in $\mathrm{mol\,L^{-1}}$, so the unit matches and the task does not. MAE is not available for the third block, which reports RMSE and $\% \log S \pm 1$, and we do not convert between them.

Limitations: what is untested, and what is not auditable. (vii) Absolute scaffold accuracy is encoder-bound but well *above* the inter-lab noise floor of §2.2, so label noise does not explain the ceiling. (viii) The latent-departure result is three-seed mean±sd (n=44): what it establishes is the *sign* and a transferable shared direction, the magnitudes ($51 \pm 2\%/72 \pm 1\%/4.1 \pm 0.5\times$) being reported but not stable across analysis configurations. The two top-two shares in play are distinct quantities: the learned-versus-reference deviation clears a smoothness-matched phase-randomised null but not a wrong-molecule one, and is measured on one grounded checkpoint rather than three seeds, whereas the isolation drift has no matched null of its own. (ix) The probe on the unconstrained control is bounded by the same reference scarcity: its crystal-term and temperature tests rest on 13 held-out solutes and cannot be enlarged, so the negative is an upper bound at $R^2 \approx 0.30$ and not a demonstration that no crystal information is present; the crystal-slope test is additionally limited by the corpus, whose empirical slope carries $-\Delta H_{\mathrm{fus}}/R$ at $r = +0.24$. The scaffold-organisation result needs no external reference, but its two partitions nearly coincide by construction, so a ratio near one is weak evidence of within-scaffold indistinguishability (§S10). (x) The closure-fidelity contrast compares *oracle-input* activity error across kernels, so $\mathcal{G}$ itself is not measured under a modern closure on either axis, and whether the *solubility* paradox survives a 2010/dsp closure is untested; both need a typed σ-profile head trained end to end through the 2010 kernel, and the solubility test a UD ∩ BigSolDB matched set. (xi) No difference in accuracy between the physics and control arms is attributable to the thermodynamic bottleneck; the confound and what survives it are stated above. (xii) The scaffold-leak guard is a property of the released stream builder and not a certified property of these runs (§2.2). The paradox contrast is one checkpoint evaluated two ways and so cannot be generated by such a leak, and DirectGNN uses no stream, but auditability is not restored by those arguments. (xiii) Several of the numbers the verdict rests on are checkable only by rerunning the released code, whose input artifacts are larger than the repository carries; the two decomposition sets are therefore deposited as row-level tables (Data Sets S1 and S2), so the sets and their margins can be recomputed without code, which does not deliver the provenance of the closure evaluations in them or of the trained arms.

4 Conclusions

Grounding a learned physical intermediate in reference values is two operations, not one, and only one of them helped here. Training the intermediate against reference values improved the model; supplying those values at prediction time did not. On the activity data where the two terms can be separated, they do not separate the same way in every chemistry: on one class of solvent, under a rule applied to every class alike, the bound on the fixed thermodynamic model's own error lies above the bound on the contribution of its inputs, which is the condition under which more accurate inputs expose a model's bias rather than remove it; on every other row set the separation is not established, at the sample sizes and the number of laboratories available. As a single number for a whole data set the comparison is an average over that set's composition, a property of the set and not of the model. And the intermediate trained through that model stopped reporting the physical quantity it is named after, on the molecules where it can be held against its reference, which is the price of reading a grounded latent as a measurement.

The general statement is a design rule with a test attached. Wherever a predictor composes a fixed physical model over a learned intermediate for which an external reference exists, that reference measures which of the two limits accuracy, and where it answers, the answer decides the remedy: a more faithful physical model, or better inputs. The answer comes chemistry by chemistry rather than once for a field, and it comes one-sided, so what the measurement returns is the chemistry on which better inputs are not the remedy—here one class of solvent, on which the standard route to a more faithful model did not clear the bar either. Whether a latent trained through an imperfect model becomes, in general, a surrogate for that model's error is the prediction this work leaves testable.

Author contributions

N. L. Polomoshnov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. A. V. Rudik: Writing – review & editing.

Conflicts of interest

There are no conflicts to declare.

Data availability

The analysis code, and those intermediate artifacts small enough to version, are available at `https://github.com/doctawho42/tgnn-solv` (commit **[hash to be supplied at submission]**) under the MIT licence, that commit being the one the reported numbers were produced from; the model checkpoints, the processed split, the full per-arm prediction files and the larger analysis artifacts will be deposited in a Zenodo archive, **[whose DOI is registered at submission]**. The data sources are public: BigSolDB 2.0[23] for solubility, the VT-2005 database for the reference σ-profiles, Ref.[29] as redistributed with Ref.[25] for the UD profiles, ThermoML for the infinite-dilution activity coefficients, the PGL 6th-edition IDAC database for Table 2, and the OPERA compilation[32] for the pKa data. Four deposited artifacts carry disclosed defects, recorded in §S9 and §S3: the seed-44 oracle per-row file is truncated, the three-seed surrogate run's per-run split provenance was not logged, one σ-stream build on disk was written against the wrong split files, and the two VT-2005-matched audit artifacts disagree on one out-of-fold estimate.

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Supporting Information Available

The Supporting Information is a separate document. Its sections, tables, figures and equations are numbered S1, S2, …, and every reference above that carries an S is to it. It contains supplementary methods; the remaining proofs; per-arm tables and decomposition robustness; the map's 59 strata in full, including the 44 the estimator cannot bound; further external ablations; supplementary mechanism results; supplementary diagnostics and broad negatives; the second-domain pKa construction; data provenance and reproducibility; the black-box probe tables; the synthetic closure-fidelity family; and the positioning against adjacent literatures. Two machine-readable tables are deposited with it: the 477-row broad IDAC set (Data Set S1, `broad_idac_set_477.csv`) and the 60-row VT-2005-matched set (Data Set S2, `vt2005_matched_set_60.csv`); their columns are listed in §S9.

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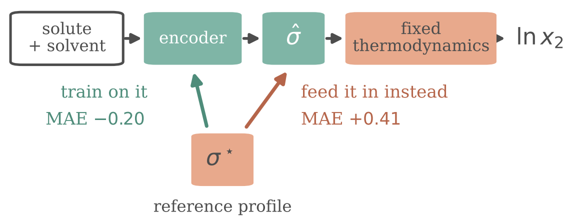
TOC Graphic



Grounding is two operations on one database, with opposite signs (solubility MAE on held-out scaffolds, three seeds).